BA 3204 — Fundamentals of MIS
Chapter 2 Global E-business and Collaboration

Learning Objectives

After studying this chapter, students should be able to answer the following questions:

- Explain what business processes are and how they are related to information systems.
- Explain how systems serve the different management groups in a business, and how systems that link the enterprise improve organizational performance.
- Explain why systems for collaboration and social business are so important, and identify the technologies they use.
- Describe the role of the information systems function in a business.

2.1 What Are Business Processes? How Are They Related to Information Systems?

To operate, a business must handle a huge amount of information about suppliers, customers, employees, invoices, and payments, and organize the work that uses this information. Information systems make it possible for firms to manage this information, make better decisions, and execute their business processes more effectively.

Business Processes

Business Process: The manner in which work is organized, coordinated, and focused to produce a valuable product or service. It is the collection of activities required to produce that product or service, supported by flows of material, information, and knowledge among participants.

A firm's performance depends heavily on how well its business processes are designed and coordinated. Well-run processes can be a source of competitive strength; poorly run ones can be a liability that slows the firm down. Every business is really a collection of business processes; some tied to a single functional area and some that cross several departments. For example, the sales and marketing function is responsible for identifying customers, and the human resource's function is responsible for hiring employees. Table 2-1 describes some typical business processes for each of the functional areas of business.

TABLE 2.1 EXAMPLES OF FUNCTIONAL BUSINESS PROCESSES

FUNCTIONAL AREA	BUSINESS PROCESS
Manufacturing and production	Assembling the product Checking for quality Producing bills of materials
Sales and marketing	Identifying customers Making customers aware of the product Selling the product
Finance and accounting	Paying creditors Creating financial statements Managing cash accounts
Human resources	Hiring employees Evaluating employees' job performance Enrolling employees in benefits plans

Fulfilling a customer order is a good example of a cross-functional process: it requires the close coordination of the sales, accounting, and manufacturing functions, with information flowing rapidly from one decision maker to another, to business partners such as delivery firms, and to the customer. Computer-based information systems make this rapid flow of information possible.

How Information Technology Improves Business Processes

Information systems improve business processes by making **work faster, easier, and more efficient**. They can **automate tasks that were previously performed manually**, such as checking customer credit, generating invoices, and preparing shipping orders.

Beyond automation, information technology can **change the flow of information itself**: it lets more people access and share information, replaces sequential steps with tasks that can be carried out simultaneously, and cuts out delays in decision making. **New technology can also change** the way a business operates and create entirely new business models. Downloading an e-book, buying a computer online, and streaming a music track are business activities that only exist because of modern information technology.

This is why analyzing business processes matters: it gives a clear picture of how a business actually works, and it is the starting point for redesigning processes to make them more efficient, more innovative, or more responsive to customers.

2.2 How Do Systems Serve the Different Management Groups in a Business, and How Do Systems That Link the Enterprise Improve Organizational Performance?

Organizations use different types of information systems because they have different departments, functions, and management levels. No single system can provide all the information an organization needs.

Information systems support major business functions such as sales and marketing, manufacturing and production, finance and accounting, and human resources. Because functional systems that operate independently of one another cannot easily share information to support cross-functional processes, many organizations have replaced them with large-scale, cross-functional systems that integrate related activities across departments.

Information systems also support different management levels:

- Operational Management: supports day-to-day activities and routine decisions.
- Middle Management: supports monitoring, control, and short- to medium-term decisions.
- Senior Management: supports strategic decisions and long-term planning.

Systems for Different Management Groups

A business firm has systems to support each of these groups or levels of management. These systems fall into two broad categories: transaction processing systems and systems for business intelligence.

Transaction Processing Systems (TPS)

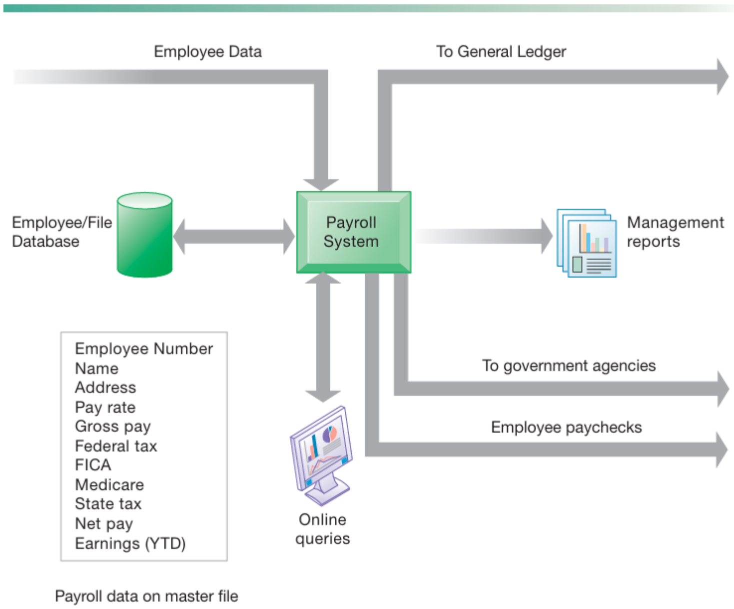
Transaction Processing System (TPS): A computerized system that performs and records the daily routine transactions necessary to conduct business, such as sales order entry, hotel reservations, payroll, employee record keeping, and shipping.

Operational managers need TPS to keep track of the elementary activities and transactions of the organization: sales, receipts, cash deposits, payroll, credit decisions, and the flow of materials in a factory. The principal purpose of a TPS is to answer routine questions (How many parts are in inventory? What happened to a customer's payment?) and to track the flow of transactions through the organization. Because of this, the information a TPS provides must be current, accurate, and easily available.

At the operational level, tasks, resources, and goals are predefined and highly structured. A credit-granting decision, for example, is made by a lower-level supervisor purely by checking whether the customer meets predefined criteria.

A payroll TPS illustrates the point: an employee time sheet is a single transaction that, once entered, updates the master file that permanently holds employee data. The system then produces paychecks and reports for management and government agencies. TPS are often so central to daily operations that even a few hours of downtime can be extremely costly and disruptive, for instance a package-delivery firm without a working tracking system, or an airline without its reservation system. TPS are also the main source of data that feeds the other systems described below, such as the general ledger, human resources, and government reporting systems.

FIGURE 2.2 A PAYROLL TPS

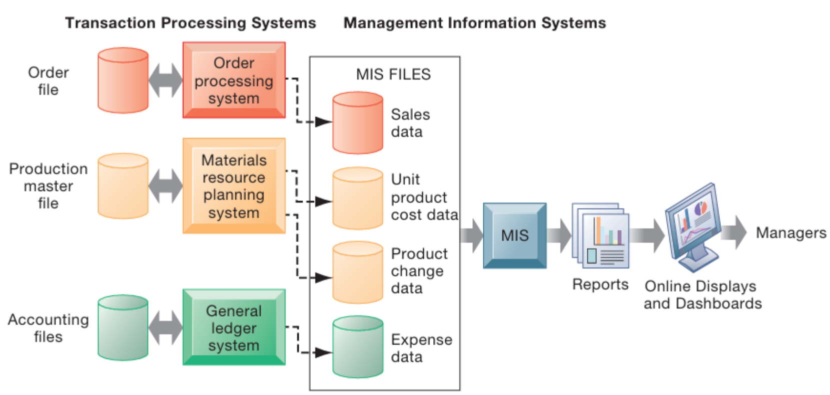


Management Information Systems (MIS)

Management Information Systems (MIS): A specific category of information system serving middle management by providing reports on the organization's current performance, used to monitor and control the business and to predict future performance.

MIS summarize and report on the company's basic operations using data supplied by TPS. This transaction-level data is compressed and typically delivered on a regular schedule (often online today) in the form of routine reports, such as comparing planned and actual sales figures for a product by region. MIS typically answer routine questions that have been specified in advance and follow a predefined procedure. They generally are not flexible and rely on simple routines, such as summaries and comparisons, rather than sophisticated mathematical models or statistics.

FIGURE 2.3 HOW MANAGEMENT INFORMATION SYSTEMS OBTAIN THEIR DATA FROM THE ORGANIZATION'S TPS

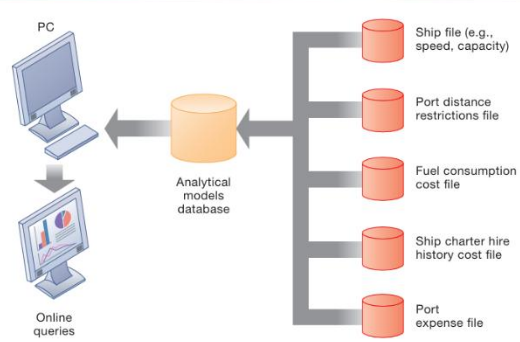


Decision-Support Systems (DSS)

Decision-Support Systems (DSS): Systems that focus on problems that are unique and rapidly changing, for which the procedure for arriving at a solution may not be fully predefined in advance.

DSS try to answer questions such as: What would happen to production schedules if sales doubled in December? What would happen to return on investment if a factory schedule slipped by six months? While DSS use internal data from TPS and MIS, they often also draw on external information, such as current stock prices or competitors' prices. They are used by "super-user" managers and business analysts who apply sophisticated analytics and models.

FIGURE 2.5 VOYAGE-ESTIMATING DECISION-SUPPORT SYSTEM

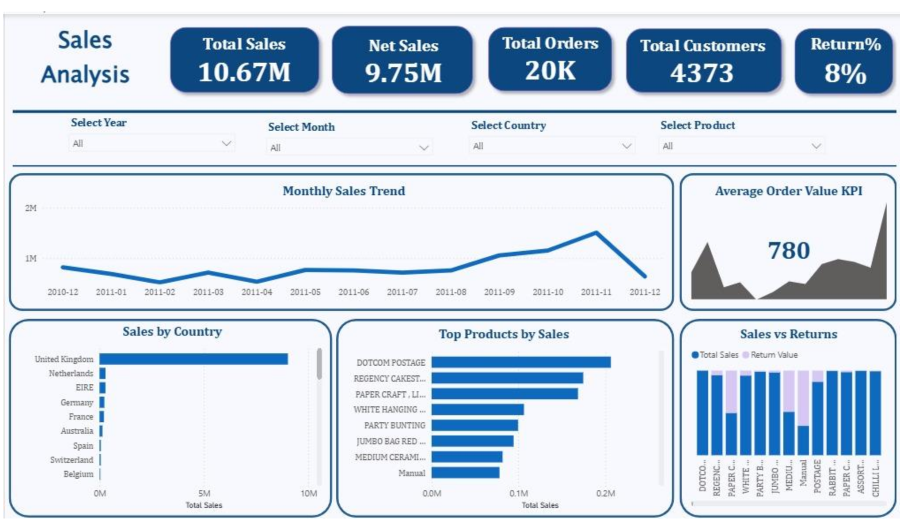


A classic example is a voyage-estimating DSS used by a global shipping company. It calculates financial details (ship and time costs, freight rates, port expenses) and technical details (cargo capacity, speed, port distances, fuel and water consumption, loading patterns) to answer questions such as which vessel should be assigned to a delivery at what rate to maximize profit, or the optimal loading pattern for a particular route. Other DSS are more data-driven: for instance, large ski resort companies analyse customer data gathered from call centres, lodging, dining, and equipment rental to determine each customer's value and target marketing more effectively.

Executive Support Systems (ESS)

Executive Support Systems (ESS): Systems that help senior management make nonroutine decisions requiring judgment, evaluation, and insight, where there is no agreed-on procedure for arriving at a solution.

Senior managers are concerned with strategic issues and long-term trends both inside the firm and in the external environment, questions such as future employment levels, long-term industry cost trends, or which products the firm should be making in five years. ESS present graphs and data drawn from many internal and external sources through an interface designed to be easy for senior managers to use, often delivered through a web-based portal. ESS are built to incorporate external events, such as new tax laws or competitor moves, while also drawing summarized information from internal MIS and DSS. They filter, compress, and track the most critical data, frequently displaying it through digital dashboards that give managers a single-screen overview of key performance indicators so they can quickly spot areas needing attention.



Systems for Linking the Enterprise

Firms typically end up with many different systems, often the product of years of organic growth and acquisitions, and face the challenge of getting them to work together as a single corporate system. This gives rise to enterprise applications.

Enterprise Applications: Systems that span functional areas, focus on executing business processes across the entire business firm, and include all levels of management. They help businesses become more flexible and productive by coordinating business processes more closely and by integrating groups of processes around efficient resource management and customer service.

There are four major types of enterprise application, each integrating a related set of functions and processes to improve the performance of the organization as a whole, and in some cases extending beyond the organization to customers, suppliers, and other business partners:

- **Enterprise Systems (also known as enterprise resource planning, or ERP, systems):** integrate business processes in manufacturing and production, finance and accounting, sales and marketing, and human resources into a single software system, so that information that used to be scattered across many systems sits in one comprehensive data repository. For example, when a customer places an order, the order data flow automatically to the warehouse (to pick and ship the goods), to the factory (to replenish stock), and to accounting (to invoice the customer), while customer service tracks the order at every step and managers use firmwide information for both daily and long-term decisions.
- **Supply Chain Management (SCM) Systems:** help firms manage relationships with suppliers, purchasing firms, distributors, and logistics companies by sharing information on orders, production, inventory levels, and delivery, so goods and services can be sourced, produced, and delivered efficiently. The objective is to get the right amount of product from source to point of consumption in the least time and at the lowest cost. SCM systems are a type of interorganizational system because they automate the flow of information across organizational boundaries.
- **Customer Relationship Management (CRM) Systems:** help firms manage relationships with their customers by coordinating the sales, marketing, and service processes that deal with customers, in order to optimize revenue, customer satisfaction, and customer retention. CRM systems help firms identify, attract, and retain their most profitable customers.
- **Knowledge Management Systems (KMS):** enable organizations to better manage processes for capturing and applying knowledge and expertise. Firms that know how to create, produce, and deliver products and services better than others hold a form of knowledge that is unique, hard to imitate, and a source of long-term strategic benefit. KMS collect relevant knowledge and experience and make it available wherever and whenever it is needed, and also link the firm to external sources of knowledge.

Intranets and Extranets

Because enterprise applications are often costly and difficult to implement, intranets and extranets serve as lighter-weight alternatives for integrating information and speeding its flow within the firm and with outside partners.

Intranet: An internal company website, built on the same technologies as the public Internet, that is accessible only to employees.

Extranet: A company website that is accessible to authorized outsiders such as vendors and suppliers, often used to coordinate the movement of supplies into the firm's production process.

E-business, E-commerce, and E-government

The systems described above are turning firms' relationships with customers, employees, suppliers, and logistics partners into digital relationships built on networks and the Internet.

E-business (Electronic Business): The use of digital technology and the Internet to execute the major business processes in the enterprise, including internal management activities and coordination with suppliers and other business partners. E-business includes e-commerce.

E-commerce (Electronic Commerce): The part of e-business concerned with the buying and selling of goods and services over the Internet, along with the activities that support those transactions, such as advertising, marketing, customer support, security, delivery, and payment.

E-government: The application of the Internet and networking technologies to digitally enable government and public-sector agencies' relationships with citizens, businesses, and other arms of government.

E-government improves the delivery of government services, makes government operations more efficient, and empowers citizens with easier access to information, such as renewing a driver's licence or applying for benefits online, as well as the ability to organize and mobilize electronically around causes.

2.3 Why Are Systems for Collaboration and Social Business Important, and What Technologies Do They Use?

Information systems alone cannot make decisions, hire or fire people, sign contracts, agree on deals, or adjust prices to the market. Beyond the systems already described, businesses also need systems that support collaboration and teamwork so that people can pull all this information together and work toward common goals.

What Is Collaboration?

Collaboration: Working with others to achieve shared and explicit goals. It focuses on task or mission accomplishment and takes place within a business, or between businesses.

Collaboration can be short-lived, lasting only minutes, or long-term; it can be one-to-one or many-to-many. It may happen in informal groups outside the formal organizational structure, or within formal teams that have been assigned a specific mission, such as "increase online sales by 10 percent." Teams are often short-lived, lasting only as long as it takes to solve the problem they were formed to address.

What Is Social Business?

Social Business: The use of social networking platforms, including public platforms and internal corporate tools, to engage employees, customers, and suppliers, allowing them to set up profiles, form groups, and follow one another's updates.

The goal of social business is to deepen interactions with groups inside and outside the firm in order to speed up and improve information sharing, innovation, and decision making. A central idea is that conversations about the firm are already happening among customers, suppliers, employees, managers, and oversight agencies, often without the firm's knowledge. If firms can tune into these conversations, they can strengthen their bonds with the people involved. This requires a high degree of information transparency, where people share opinions and facts directly rather than through executives or intermediaries.

If achieved, this kind of environment can drive operational efficiencies, spur innovation, and speed up decision making, for example by letting product designers learn directly from real-time customer feedback, or letting employees use internal and external social connections to capture new knowledge and solve problems more efficiently.

Building a Collaborative Culture and Business Processes

Collaboration will not happen on its own without a supportive culture and supportive business processes. Many firms historically operated as "command and control" organizations, where senior leaders decided what mattered and lower-level employees carried out those decisions without much question, with no reward for teamwork and no expectation of horizontal communication between work groups; problems between groups were left for the bosses to resolve.

A collaborative culture looks different. Senior managers remain accountable for results but rely on teams at every level to devise, create, and build the policies, products, designs, processes, and systems needed to get there. Teams, and individuals within them, are rewarded for team performance, and the role of middle managers shifts toward building teams, coordinating their work, and monitoring performance. Crucially, a genuinely collaborative culture is one where senior management treats collaboration and teamwork as vital and models it at the senior level, not just further down the organization.

Tools and Technologies for Collaboration and Social Business

A collaborative, team-oriented culture only produces results when it is backed by the right information systems. There are now hundreds of tools that support the fact that employees, customers, suppliers, and managers depend on one another more than ever to get work done. Some of these are expensive; many are available for free or for a modest fee.

- **E-mail and Instant Messaging (IM):** Long-standing communication and collaboration tools that run on computers, mobile phones, and other wireless devices, supporting file sharing as well as messaging, with many systems allowing real-time conversations among multiple participants at once. In recent years usage has shifted toward messaging apps and social media as the preferred channel.
- **Wikis:** Websites that let users add and edit text and graphics without needing web development or programming skills. Wikipedia, the largest collaboratively edited reference project in the world, is the best-known example. Firms use wikis internally and externally to store and share knowledge, for instance as a knowledge base for customers and outside developers, replacing the inefficient practice of repeatedly answering the same questions on forums.
- **Virtual Worlds:** Online 3-D environments populated by "residents" represented by avatars, such as Second Life. Companies use these environments for meetings, interviews, guest-speaker events, and employee training, with participants interacting through gestures, chat, and voice communication.
- **Collaboration and Social Business Platforms:** Multifunction software suites that support collaboration among teams working from different locations, including Internet-based audio and video conferencing, cloud collaboration services, corporate collaboration systems, and enterprise social networking tools that let employees set up profiles, form groups, and follow updates within the firm.
- **Virtual Meeting Systems:** Videoconferencing and web-conferencing technologies that reduce travel costs and let people in different locations meet and collaborate; companies use them for product briefings, training courses, strategy sessions, and other meetings.

2.4 What Is the Role of the Information Systems Function in a Business?

Businesses need many different kinds of information systems to operate, but running and maintaining those systems, keeping the hardware, software, and other technologies up to date and working properly, requires a dedicated information systems function. End users manage their systems from a business standpoint, while the technology itself is managed by IS specialists.

The Information Systems Department

In all but the smallest firms, the information systems department is the formal organizational unit responsible for information technology services, including maintaining the hardware, software, data storage, and networks that make up the firm's IT infrastructure. This department is staffed by specialists such as:

- **Programmers:** highly trained technical specialists who write the software instructions for computers.
- **Systems Analysts:** the principal link between the information systems group and the rest of the organization, whose job is to translate business problems and requirements into information requirements and systems.
- **Information Systems Managers:** leaders of teams of programmers and analysts, as well as project managers, facility managers, telecommunications managers, database specialists, and managers of computer operations and data entry staff.

External specialists such as hardware vendors, software firms, and consultants also regularly take part in the day-to-day operation and long-term planning of a firm's information systems.

The information systems department is usually headed by a Chief Information Officer (CIO), a senior manager who oversees the firm's use of information technology and is expected to combine a strong business background with information systems expertise, playing a leadership role in aligning technology with business strategy. Larger firms may also have several related senior roles that work closely with the CIO:

- **Chief Security Officer (CSO), sometimes Chief Information Security Officer (CISO):** in charge of information systems security, responsible for enforcing the firm's security policy, training staff, tracking threats, and maintaining security tools.
- **Chief Privacy Officer (CPO):** responsible for ensuring the company complies with data privacy laws, a role created as safeguarding personal data became increasingly important.
- **Chief Knowledge Officer (CKO):** responsible for the firm's knowledge management program, helping to design ways of finding new knowledge or making better use of existing knowledge.
- **Chief Data Officer (CDO):** responsible for enterprise-wide governance and use of information, ensuring the firm collects the right data, deploys the right technology to analyse it, and uses the results to support business decisions.

End users, the employees outside the IS group for whom systems are built, are playing an increasingly large role in the design and development of information systems. Over time, the information systems department has also shifted away from being mostly programmers performing narrow technical tasks, toward a broader mix of systems analysts and network specialists who act as change agents, proposing new business strategies and new information-based products and services, and coordinating both the technology and the organizational change that goes with it.

Careers in information systems and MIS continue to grow faster than the average for the economy as a whole and tend to be well compensated relative to other occupations, with particularly strong demand in areas such as systems analysis, database administration, information security, and software engineering, and steady demand for IS/IT management roles as firms rely more heavily on the Internet for computing and communication.

Organizing the Information Systems Function

There is no single way to organize the IT function; it depends on the size and nature of the firm. A very small company may not have a formal information systems group at all, relying instead on a single employee or outside consultants to keep networks and applications running. Larger companies typically have a dedicated information systems department, organized in whichever way best suits the firm's structure and priorities.

IT Governance: The strategy and policies for using information technology within an organization, specifying the decision rights and accountability framework needed to ensure that IT use supports the organization's strategy and objectives.

Decisions covered by IT governance include how centralized the information systems function should be, what decisions are needed to ensure effective use of IT and a strong return on IT investment, who should make those decisions, and how they will be made and monitored. Firms with strong IT governance have thought through clear answers to each of these questions.

Review Question:

1. What are business processes? How are they related to information systems?
2. How do systems serve the different management groups in a business, and how do systems that link the enterprise improve organizational performance?
3. What is the role of the information systems function in a business?